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RCED v5.1 — Comparison of three regimes

A 20-year projection of the planetary boundaries, emissions, energy consumed and the Good Life Index, under three regimes: RCED v5.1, social-democratic and free market.

RCED v5.1 Social-democratic Free market

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Illustrative parametric simulation, calibrated to the model's documented endpoints (F1, F6, F7). The social-democratic regime follows the model's § 5.3 scenario (private property + public investment + regulation). Read as a comparison of dynamics, not as an empirical forecast.

SIMULATION REPORT · THREE REGIMES

1. Purpose & method

Three regimes share the same automation base (r ≈ 4%/yr, constraint F1) but differ in how gains are used and in the property structure. The simulation compares their 20-year dynamics across four dimensions: planetary boundaries, net emissions, final energy and the Good Life Index (IVB).

- **RCED v5.1** — use-property, allocation F1, 9 boundaries (Layers 12-14 + 12.A/12.B), DSS-PS-EE governance (§ 7 bis).
- **Social-democratic (§ 5.3)** — private property retained + public investment + regulation. Reaches the viability zone by "patching" each boundary onto an intact growth engine.
- **Free market** — gains reinjected into material growth, relative decoupling, rebound effect, no structural reform.

NATURE OF THE EXERCISE

An **illustrative** parametric model, calibrated to reproduce exactly the RCED's documented endpoints at year 20. The social-democratic and market trajectories rest on explicit assumptions (§ 7). Read as a comparison of *dynamics*.

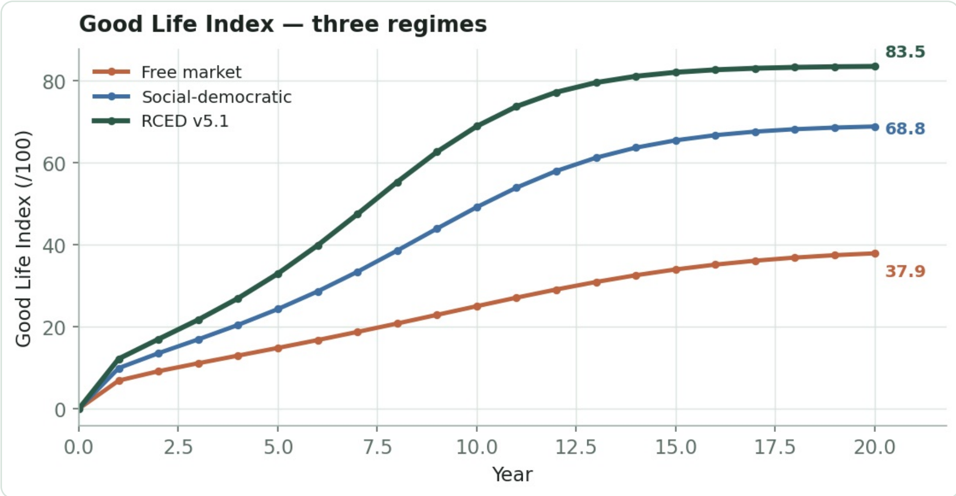
2. Key results — Year 20

Indicator (Y20)	RCED v5.1	Social-democratic	Free market
Good Life Index (IVB)	83.5	68.8	37.9
Social foundation	80.9	75.0	57.0
Ecological ceiling	86.2	63.2	25.2
Net emissions (index)	32	40	80
Final energy (index)	77	96	112
Planetary boundaries respected	9/9	9/9	3/9
Gini coefficient	0.185	0.21	0.38
Working hours / week	28.5	34.0	41.5
GDP (internal index)	280	309	333

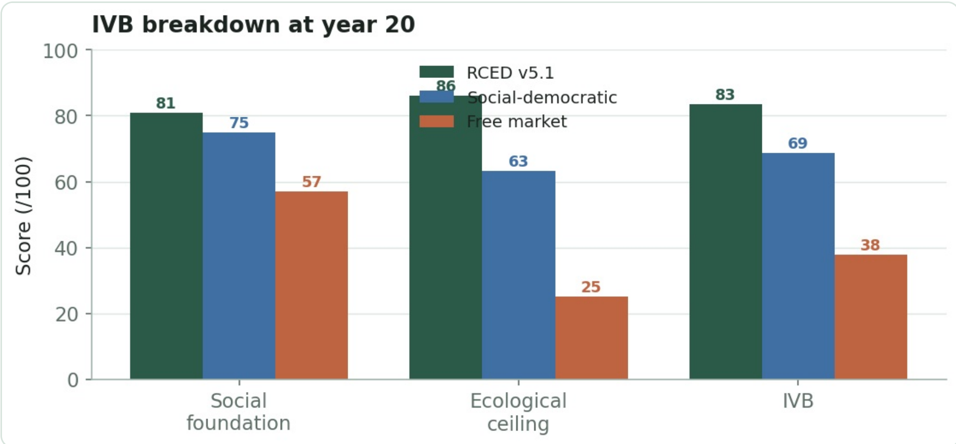
IVB ordering: RCED 83.5 > social-democratic 68.8 > market 37.9. Social democracy reaches all 9 boundaries (9/9) and stays strong on the social side, but its ecological ceiling lags (63.2 vs 86.2): material sufficiency drops off, with the highest GDP after the market.

3. Good Life Index

The three regimes rank clearly. RCED converges to **83.5**, social democracy to **68.8**, the market to **37.9**. The RCED ↔ social- democratic gap comes not from the social side (social foundation 75 vs 81, close) but from the ecological ceiling (63 vs 86): material sufficiency.



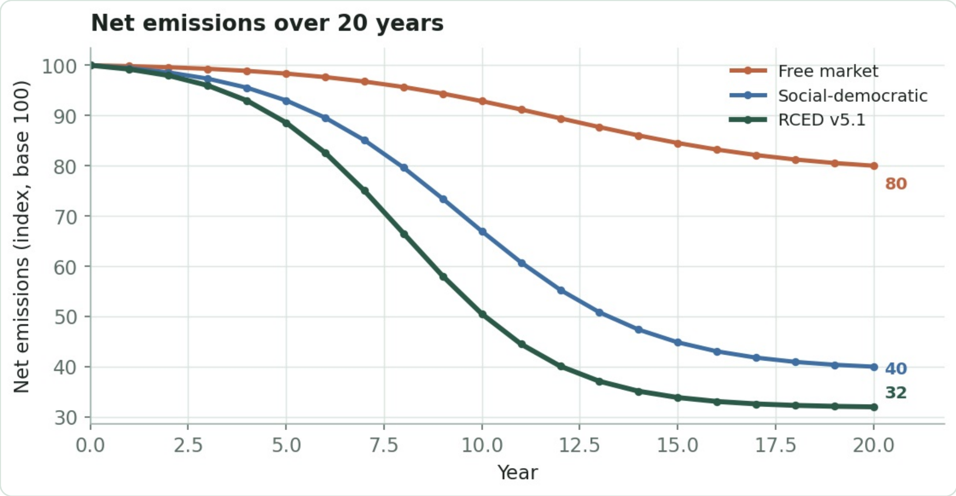
Good Life Index over 20 years, three regimes.



Breakdown at year 20: social democracy holds the social foundation but not the ecological ceiling.

4. Net emissions

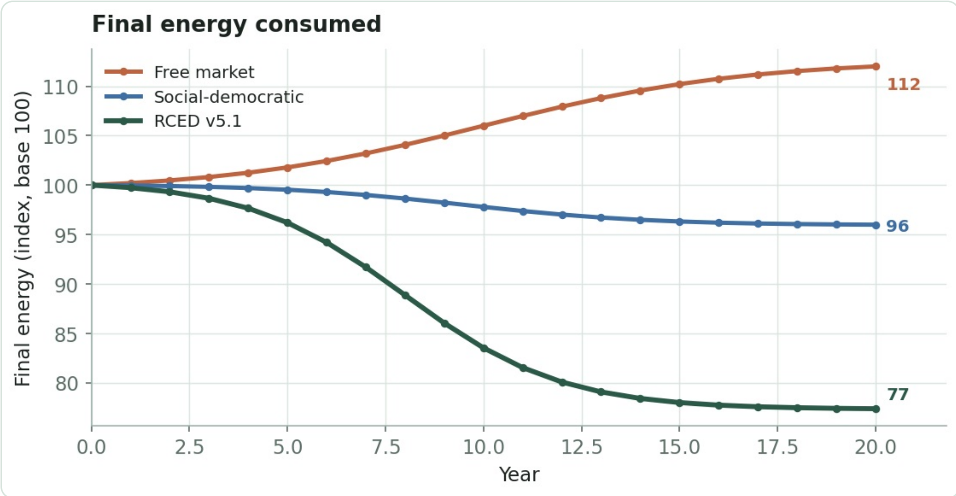
On climate, social democracy and RCED converge: **−60%** (index 40) versus **−68%** (index 32). Regulation and public investment do most of the climate work. The free market lags far behind at **−20%** (index 80).



Net emissions (index, base 100). Climate is where social democracy comes closest to RCED.

5. Final energy consumed

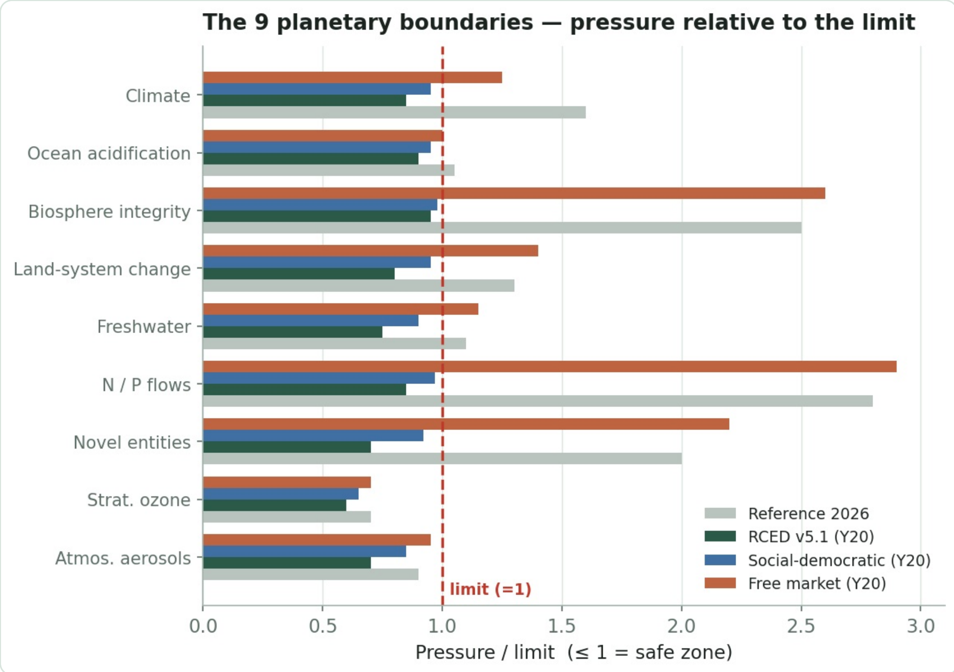
Here the structural difference shows. RCED **cuts** its final consumption (index 77, −22.6%). Social democracy barely stabilises it (index 96, −4%): regulation improves efficiency, but the intact growth engine keeps the volume up. The market sees it **rise** (index 112, rebound effect).



Final energy (index, base 100): structural sufficiency (RCED) vs efficiency without sufficiency (social democracy) vs rebound (market).

6. Planetary boundaries

Social democracy and RCED both reach the safe zone (**9/9** and **9/9**), confirming the § 5.3 finding: regulation can "patch" the boundaries. But social democracy **hugs** them (pressures near 1), where the RCED keeps a structural margin. The market respects only 3/9 and worsens several.



Pressure relative to the limit (≤ 1 = safe zone). Social democracy (blue) hugs the limit; RCED (green) keeps a margin.

Boundaries respected over time: 9/9 (RCED) and 9/9 (social democracy) vs 3/9 (market).

7. Robustness, durability & limits of the exercise

THE REAL RCED ↔ SOCIAL-DEMOCRATIC GAP

The 20-year simulation **understates** the RCED's advantage, because it does not capture long-run durability. Social democracy holds the zone through policies *grafted* onto an intact growth-and-rent engine: thin margin (pressures ~ 1), a bet on absolute decoupling, reversibility with political cycles. RCED *structurally* removes the incentive to rent and overproduce (use-property), hence a wider margin and a more robust hold beyond the simulated horizon. This is the model's central open question: regulated green growth vs structural removal of the material engine.

LIMITS OF THE SIMULATION

- **Parametric model calibrated** to the RCED's endpoints — those RCED endpoints are posed, not proven.
- Durability beyond 20 years (robustness to shocks, political reversibility) is not modelled: it favours RCED but is not quantified here.
- A market with a strong carbon tax would improve climate but stay weak on biosphere / N-P / plastics.
- No stochastic dynamics and no full macro-monetary closure (GDP is indexed). IVB weights follow the model's conventions.

A. Assumptions & equations

Transition curves (rescaled logistic)

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s(t) = [L(t)−L(0)]/[L(20)−L(0)], L(x)=1/(1+e^(−k(x−t0)))
RCED: t0=8, k=0.50 · Social-dem.: t0=9.5, k=0.42 · Market: t0=12, k=0.32
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Year-20 targets (sub-scores /100)

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RCED : soc.{72; 81.5; 81; 90} eco.{climate 85; boundaries 100; sufficiency 75.3}
Soc-dem: soc.{70; 79; 65; 88} eco.{climate 80; boundaries 90; sufficiency 35}
Market: soc.{66; 62; 33; 78} eco.{climate 40; boundaries 40; sufficiency 10}
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RCED targets = documented values (Fsoc 80.9 ; Feco 86.2 ; IVB 83.5).

Good Life Index (F7) & viability (F6)

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IVB = √( F_soc × F_eco ), F_soc=(GNH·(1−Gini)·100·Time·Needs)^(1/4), F_eco=(Climate·Viab·Sufficiency)^(1/3)
respected(t) = Σ_i 1[ pressure_i(t) ≤ limit_i ], i=1..9
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